

ATTENTION!

Installation of this coil **MUST** be performed by a GE field engineer. Please contact your field engineer to arrange for installation.

Attempts to perform installation of this coil without proper system configuration will result in system errors.

GE FE: Installation notes are attached to this document.

Attention GE Field Engineer:

The installation process for this coil requires 4 steps:

- 1) If needed, upgrade system software to the appropriate version.
- 2) If needed, install required patches.
- 3) Add configurations.
- 4) Run SNR QA tool.

The coil should not be turned over to the customer until these 4 steps have been completed.

IMPORTANT NOTE 1: THIS COIL MUST BE USED ONLY AFTER THE PROPER UPGRADES HAVE BEEN INSTALLED. PLUGGING IN THE COIL IN A SYSTEM WITHOUT THE UPGRADE MAY RESULT IN PROBLEMS IN THE SYSTEM OR THE COIL!!!!!!

IMPORTANT NOTE 2: This document consists of Software installation procedures for the 12.X, G3 and E2 systems. Please follow appropriate instructions.

THIS IS ESPECIALLY SO IN SITES WHERE SYSTEMS ARE INSTALLED IN ADJACENT ROOMS OR IN THE VICINITY OF EACH OTHER AND THE COILS HAVE A HIGH PROBABILITY BEING TRANSPORTED FROM ONE SYSTEM TO ANOTHER.

The installation procedure is attached.

Note: This following instructions are only for the for the 12.X systems with Connector type (MCAT 3334TP [HD], M1385AT [LEGACY]). For E2 and G3 systems, please go further in the documentation.

1. Confirm that the site has been upgraded to Excite HD 12.x M4 (12.0_M4_0520.a) or higher software. If the site has not yet been upgraded, perform the upgrade (available via FMI 60698) before continuing with installation of this coil.
2. If the site is running M4 software, install patch **12.0 M4 20-PC-1** from the boot mode and restart Signa. If the patch has not been pushed to the site via broadband, it is available via the patch download quickplace or CD.

Note: A good way to download the patch from the CD is via the Guided Install Tool – FE Mode – name (root), password (operator) – Patches – CD ROM – Install.

IMPORTANT!!! : FOR 3T94 MAGNETS, PLEASE REFER TO THE APPENDIX BELOW FOR JUMPER SETTINGS.

3. Remove any coil names beginning with “HDNV” or “8NV” from the site list using the configuration file manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager.
4. Detailed installation procedures are available in the README.txt in the CD accompanying the coil – please refer to those before doing the actual installation.
5. Add the following coil names to the site list using the configuration file manager. Restart Signa so that the new coil configuration names take effect.

Coil Configuration Names for the HD Connector
HDNV Array
HDNV Head
HDNV Upchest
HDNV Spectro
HDNV ArrayServ
HDNV HeadServ
HDNV SpectroServ

Coil Configuration Names for the Legacy Connector
8NV Array
8NV Head
8NV Upchest
8NV Spectro
8NV ArrayServ
8NV HeadServ
8NV SpectroServ

6. Confirm proper operation of the coil by running the “surface coil SNR test.”
Refer to: Service Methods CD; Troubleshooting; Image quality troubleshooting
tools; Surface coils SNR test.
7. Add the coil to product locator under p/n 2414390.

Note: The following instructions are only for the E2 systems. These have Connector type M1385AT [LEGACY]. G3 procedure is given in the next section.

1. The upgrade number contained in this CD is E2 M4 Update **17-PB-2**

Note: A good way to download the patch from the CD is via the Guided Install Tool – FE Mode – name (root), password (operator) – Patches – CD ROM – Install.

The README.txt information is displayed when you click the Info button.

2. Before doing this update, please go through the README.txt file in the CD – it contains important installation information.
3. **IMPORTANT!! : PLEASE MAKE a save INFO BEFORE INSTALLING THIS UPGRADE (PATCH) – this is described in detail in the Install instructions in the README.txt file.**

Failure to do this may lead to a system crash at start-up and a full load from cold!

IMPORTANT!!! : FOR 3T94 MAGNETS, PLEASE REFER TO THE APPENDIX BELOW FOR JUMPER SETTINGS.

4. This upgrade, in Addition to the 3T NV compatibility, takes care of a few other issues which are described below:

Note: For all the changes described below, no individual steps have to be performed. All the updates will be performed by the CD software installation.

4a. CHANGE 1: Two Coil Config Entries for 3CH Shoulder Coil

At M4 release, the 3CH Shoulder Coil had two entries in the Coil Config file, causing the system to crash during load-from-cold. Update 00-PA was created to address this, and must be installed during the load-from-cold process. Because update 17 makes further modifications to the Coil Config files, it conflicts with update 00.

This change is included in updates

00-PA

17-PA, PB, and PC

4b. CHANGE 2 - Power Monitor Trips with CTL Coils

With CTL coils, there seems to be inconsistencies in TG setting leading to Power Monitor trips when TG is set too high and poor IQ when TG is set too low.

This change is included in updates
17-PA, PB, and PC

4c. CHANGE 3 - csa and cstun do not work

Fixes the csa cstun problem ChemSat.e.

This change is included in updates
17-PA, PB, and PC

4d. CHANGE-4: Image Quality using new CTL Coil

After new CTL coil installation, image quality for L-spine FSE T1/T1flair was unsatisfactory.

This change is included in updates
17-PA, PB, and PC

4e. CHANGE 5 - Applied B-Value for DWI with DSE.

The applied b-value for DWI/DTI+DSE was off by about 10% from what user had prescribed.

This change is included in updates
05-PA

17-PA, PB, and PC

4f. CHANGE 6 - ACGD-Related Banding in FSE Applications

This update fixes ACGD-related ghosting and banding, as well as implement enhancements to phase correction and error checks for Maxwell term compensation.

This change is included in updates
17-PB and PC

4g. CHANGE 7 – Faults with 8 Channel CTL coils.

Software Update 17-PA contains the coil configurations necessary to support all the 8 Channel CTL coil configurations. But they have an Open Circuit Trip

Limit of 400 mA rather than 350 mA. This leads to spurious faults. This change corrects all the 8USCTL... coil entries.
It is necessary to use the Config File Manager to remove and re-install these coil configurations.

This change is included in updates
17-PB and PC

4h. CHANGE 8 – SNR Test for 8 Channel CTL Coils.

Software Update 17-PA contained the coil configurations necessary to support all the 8 Channel CTL coil configurations. But it was missing the revisions to the SNR Test Tool to calibrate them properly.

This change is included in updates
17-PC

4i. CHANGE 9 - Support for 8 Channel NV Array Coil from USAI

This change provides support for the 8 Channel NV Array Coil from USAI on scanners running the G3.0 M4A software. The change includes 7 new coil configuration entries in the Coil Config file, 6 new protocols for head and neck, and a new version of the run_snr_for_multicoil_dicom tool to support the coil.

This change is included in updates
17-PC

4j. CHANGE 10 - Setup Protocols did not run with Software Updates

When installing or removing a Software Update, the setupProtocols script would not execute, and it would have been necessary for the FE to manually run it in order for the new protocols to be seen on the user interface. This change fixes the setupProtocols script and executes it automatically at Update install or removal.

This change is included in updates
17-PC

5. Follow the CD installation instructions described in the README.txt file.

6. Do a RESTORE INFO from the MOD.
7. Use the Coil Config Manager to **remove (if present)** and add the following coil config entries:

8USCTLTOP
8USCTLMID
8USCTLBOT
8USCTL123
8USCTL234
8USCTL345
8USCTL456
8USCTL12
8USCTL23
8USCTL34
8USCTL45
8USCTL56
8USCTL123service
8USCTL456service
8NVArray
8NVHead
8NVSpectro
8NVUpChest
8NVArrayServ
8NVHeadServ
8NVSpectroServ

8. Restart Signa so that the new coil configuration names take effect.
9. Confirm proper operation of the coil by running the “surface coil SNR test.” Refer to: Service Methods CD; Troubleshooting; Image quality troubleshooting tools; Surface coils SNR test.
10. Add the coil to product locator under p/n 2414390.

**Note: This following document is only for the for the G3 systems.
These have Connector type M1385AT [LEGACY].**

1. The upgrade number contained in this CD is **G3.0 M4A Update 31-PF-5**

Note: A good way to download the patch from the CD is via the Guided Install Tool – FE Mode – name (root), password (operator) – Patches – CD ROM – Install.

The README.txt information is displayed when you click the Info button.

2. Before doing this update, please go through the **README.txt** file in the CD – it contains important installation information.
3. This upgrade, in Addition to the 3T NV compatibility, takes care of a few other issues which are described below:

Note: For all the changes described below, no individual steps have to be performed. All the updates will be performed by the CD software installation.

3a. CHANGE 1: Correction for 4 Channel NV Array Coil from USAI

This change corrects the multicoil bias for the 4 Channel NV Array Coil from USAI.

This change is included in updates

08-PA

25-PD

31-PD and PF

3b. CHANGE 2 - Support for 3 Channel Shoulder Coil

This change adds support for the 3 Channel Shoulder Coil.

This change is included in updates

25-PD

31-PD and PF

3c. CHANGE 3 - Support for 8 Channel CTL Coil.

This change adds support for the 8 Channel CTL Coil. The change includes 14 new coil configuration series in the Coil Config file.

This change is included in updates

25-PD

31-PD and PF

3d. CHANGE-4: SNR Tool for 8 Channel CTL Coil

This change completes support for the 8 Channel CTL Coil on scanners running the G3.0 M4A software. The change includes 3 new protocols for abdomen, chest, and neck, and a new version of the run_snr_for_multicoil_dicom tool to support the coil.

This change is included in updates
31-PD and PF

3e. CHANGE 5 - Support for 8 Channel NV Array Coil from USAI

This change provides support for the 8 Channel NV Array Coil from USAI on scanners running the G3.0 M4A software. The change includes 7 new coil configuration entries in the Coil Config file, 6 new protocols for head and neck, and a new version of the run_snr_for_multicoil_dicom tool to support the coil.

This change is included in updates
31-PF

3f. CHANGE 6 - setupProtocols did not run with Update 31-PD

When installing or removing Software Update 31-PD, the setupProtocols script did not execute, and it was necessary for the FE to manually run it in order for the new protocols to be seen on the user interface. This change fixes the setupProtocols script and executes it automatically at Update install or removal.

This change is included in updates
31-PF

4. Follow the CD installation instructions described in the README.txt file
5. Add the following coil names to the site list using the configuration file manager.

Use the Coil Config Manager to remove (if present) and add the following coil config entries:

**US4NVHead
US4NVFULL
3US SHOULDER
8USCTLTOP
8USCTLMID
8USCTLBOT
8USCTL123
8USCTL234
8USCTL345**

8USCTL456
8USCTL12
8USCTL23
8USCTL34
8USCTL45
8USCTL56
8USCTL123service
8USCTL456service
8NVArray
8NVHead
8NVSpectro
8NVUpChest
8NVArrayServ
8NVHeadServ
8NVSpectroServ

6. Restart Signa so that the new coil configuration names take effect.
7. Confirm proper operation of the coil by running the “surface coil SNR test.”
Refer to: Service Methods CD; Troubleshooting; Image quality troubleshooting tools; Surface coils SNR test.
8. Add the coil to product locator under p/n 2414390.

APPENDIX FOR JUMPERS IN 3T94 SYSTEMS [FOR THE 12.X AND E2 SYSTEMS]

Illustration 1: Please refer to the instructions for the jumpers settings in this illustration.

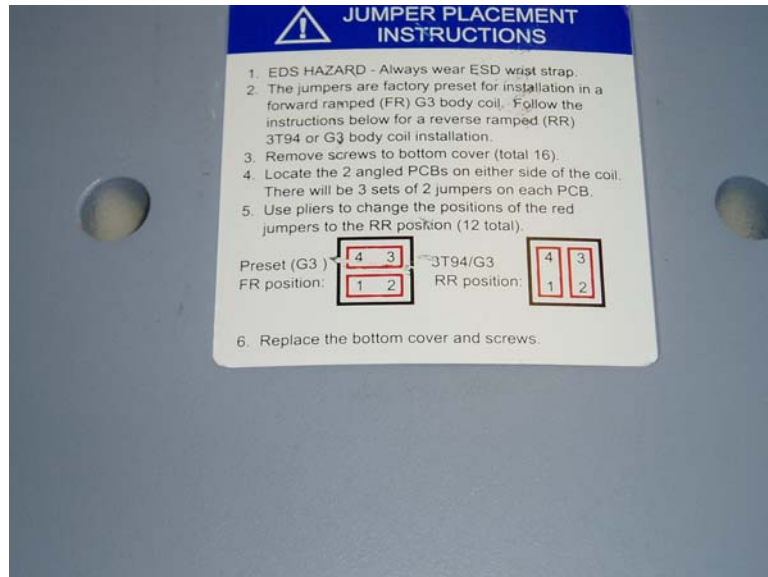


Illustration 2: The 6 jumpers are shown in the circles in the illustration

Jumpers

